(a) Scene fixture

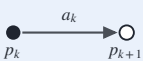
scene_fixture.json — state s_0

p0	(1, 3)	*agent (point)*
g	(9, 3)	*goal*
A	[3.4, 4.6]×[1.8, 4.2]	*Barrier*
B	c (6.6, 4.7), r 0.55	*Pier*

Task — *from the same fixture*

Reach g; avoid Barrier A and Pier B.

Rollout model

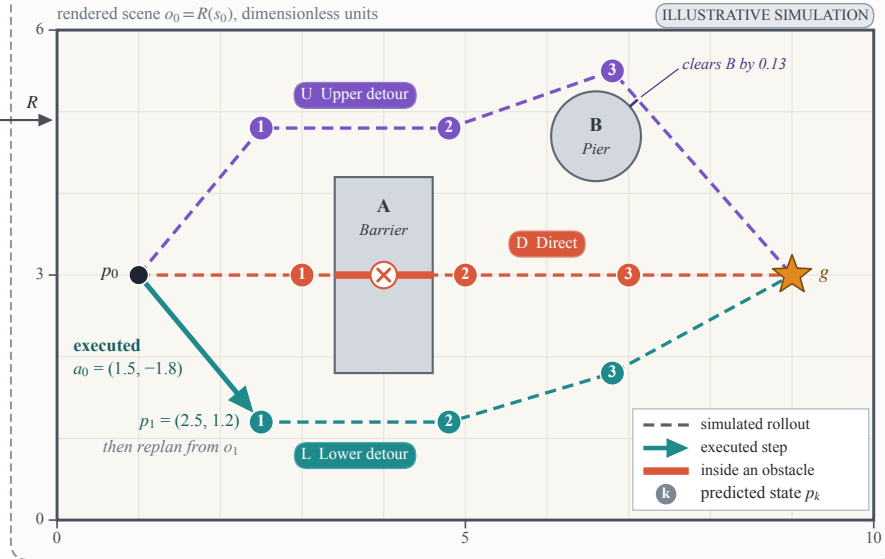


$p_{k+1} = p_k + \Delta t\, a_k$

$\Delta t = 1$ $H = 4$ $\|a_k\|_2 \le 3.5$

planner reads s_0 directly; no learned perception

(b) Scene and candidate rollouts



(c) Swept-segment test

D, segment $p_1 \to p_2$



✓ endpoints p_1, p_2 are outside A

✗ **inside A for $t \in [0.20, 0.80]$: $C = 1$**

(d) Cost and selection

$J = \|p_4 - g\|_2^2 + 0.1\,E + 100\,C, \quad E = \Sigma_k \|a_k\|_2^2$

$C = 1$ if any swept segment collides

D Direct — ✗ collides

k	0	1	2	3
a_x	2	2	2	2
a_y	0	0	0	0

$\|p_4 - g\|_2^2$		0
$0.1\,E$	$E = 16.00$	1.600
$100\,C$	$C = 1$	**100**
J		**101.600**

U Upper detour — ✓ free

k	0	1	2	3
a_x	1.5	2.3	2	2.2
a_y	1.8	0	0.7	−2.5

$\|p_4 - g\|_2^2$		0
$0.1\,E$	$E = 26.36$	2.636
$100\,C$	$C = 0$	0
J		**2.636**

L Lower detour — ✓ selected

k	0	1	2	3
a_x	1.5	2.3	2	2.2
a_y	−1.8	0	0.6	1.2

$\|p_4 - g\|_2^2$		0
$0.1\,E$	$E = 21.42$	2.142
$100\,C$	$C = 0$	0
J *minimum*		**2.142**